DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MAJOR PROJECT REPORT

AI-POWERED MULTI-AGENT COLLEGE ADMISSION ASSISTANT

A Project Report Submitted in Partial Fulfillment of the Requirements
for the Award of the Degree of
BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE AND ENGINEERING (GENERATIVE AI)

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Approved by AICTE | Accredited with NAAC 'A+' Grade | Affiliated to State University
Andhra Pradesh, India — Academic Year 2025-2026

CERTIFICATE OF APPROVAL

This is to certify that the project report entitled **'AI-POWERED MULTI-AGENT COLLEGE ADMISSION ASSISTANT'** is a bonafide record of work carried out by **[CANDIDATE NAME]** (Roll No: **[REGISTER NUMBER]**) in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** from **[INSTITUTION NAME]** during the academic year 2025-2026.

The results embodied in this report have been verified and found satisfactory under our supervision. The work has not been submitted to any other University or Institute for the award of any degree or diploma.

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CANDIDATE DECLARATION

I hereby declare that this project report entitled **'AI-Powered Multi-Agent College Admission Assistant'** is an authentic record of my own research and engineering work undertaken under the guidance of [GUIDE NAME], Department of Computer Science and Engineering. I confirm that all literature, code libraries, and empirical benchmarks used in this report have been appropriately cited and acknowledged.

Date: _____
Place: Andhra Pradesh, India

Signature of Candidate

ACKNOWLEDGEMENTS

I express my deepest gratitude to our institution's leadership, Principal, and Head of Department for fostering an academic environment conducive to practical engineering research in Generative AI. Special thanks to our faculty project supervisor for continuous mentorship and critical evaluations of the LangGraph multi-agent orchestration architecture.

I also extend my sincere appreciation to the faculty reviewers, peers, and contributors whose collaborative feedback helped test and calibrate our Andhra Pradesh cutoff algorithms and fee models.

ABSTRACT

Engineering college admissions in India require navigating complex, high-stakes decisions characterized by informational opacity and data fragmentation. Students qualifying competitive entrance examinations (e.g. AP EAPCET, JEE) face disparate data sources concerning category-specific closing ranks (OC, BC, SC, ST, EWS), tuition fees, hostel expenditures, and accredited placement outcomes. Commercial search portals frequently suffer from commercial advertising bias or ungrounded generative hallucinations.

To resolve this, we present the **AI-Powered Multi-Agent College Admission Assistant**, an autonomous multi-agent platform engineered with **LangGraph** and **FastAPI**. The platform coordinates six specialized AI agents: (1) *College Finder Agent*, (2) *Admission Eligibility Agent*, (3) *Fee Analysis Agent*, (4) *College Comparison Agent*, (5) *Admission Guidance Agent*, and (6) *AI Admission Coordinator*. Using stateful graph routing, the system evaluates candidate credentials against verified benchmarks, categorizing colleges into **Reach**, **Match**, and **Safer** options without misleading admission guarantees.

Built on React 18, Tailwind CSS, Recharts, and Node.js Express with dual cloud and embedded database resilience, the system is calibrated with verified datasets from premier Andhra Pradesh engineering institutions (AUCE, JNTUK, SVUCE, VRSEC, GVPCE, RVRJC, MBU, KLU) and models state welfare schemes including Jagananna Vidya Deevena (JVD).

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1. INTRODUCTION & PROBLEM CONTEXT

The transition from secondary education to professional collegiate engineering programs is a pivotal career juncture for Indian students. In Andhra Pradesh alone, upwards of 300,000 candidates appear annually for the AP EAPCET. Despite the availability of official seat matrices, students frequently make suboptimal choices due to the absence of unified analytical tools. Students must evaluate colleges across multiple dimensions: branch cutoffs, geographic location, financial feasibility (tuition, hostel, mess), and campus placement records.

Traditional solutions consist either of raw, unindexed government PDF bulletins or commercial search aggregators. The latter often suffer from sponsored prioritization and lack personalized reasoning. While modern Generative AI and Large Language Models (LLMs) offer conversational capabilities, naive single-prompt implementations frequently suffer from hallucinations—fabricating cutoff scores or irresponsibly guaranteeing admission.

2. PROPOSED MULTI-AGENT ARCHITECTURE

To overcome the limitations of monolithic AI, this project adopts a decoupled, multi-agent architecture utilizing **LangGraph**. Each agent operates with an explicit, segregated mandate:

- **Agent 1: College Finder Agent:** Scans institutional records and filters colleges by branch (CSE, AI&DS;, ECE, IT), district (Visakhapatnam, Kakinada, Vijayawada, Guntur, Tirupati), and campus amenities.
- **Agent 2: Admission Eligibility Agent:** Performs mathematical and percentile comparisons between candidate entrance ranks and historical closing ranks across reservation categories (OC, BC, SC, ST, EWS). Classifies options into Reach, Match, and Safer categories without guarantee.
- **Agent 3: Fee Analysis Agent:** Calculates comprehensive annual cost of attendance (Tuition + Room + Board + Amenities). Models eligibility for Andhra Pradesh Jagananna Vidya Deevena (JVD) 100% tuition reimbursement.
- **Agent 4: College Comparison Agent:** Generates comprehensive side-by-side matrices comparing up to 4 colleges across placements (highest and average LPA), NAAC accreditation, NIRF ranking, and synthesizes balanced pros and cons.
- **Agent 5: Admission Guidance Agent:** Outlines the 5-stage APSCHE web counselling roadmap, certificate verification protocols at Help Line Centers (HLC), and official state portal access links.
- **Agent 6: AI Admission Coordinator:** Orchestrates the LangGraph state machine, gathers outputs from specialist agents, and compiles a coherent, personalized admission recommendation plan.

3. SYSTEM ARCHITECTURE & TECHNOLOGY STACK

Component Layer	Technologies Deployed	Key Responsibilities
Frontend UI / UX	React.js 18, Vite, Tailwind CSS, Recharts, Lucide	12 interactive pages, charts, responsive ed-tech theme
Backend REST API	Node.js, Express.js, JWT, Bcryptjs, Mongoose	Auth, student profiles, college CRUD, AI proxy bridge
AI Multi-Agent Service	Python 3.14, FastAPI, LangGraph, OpenAI API	StateGraph orchestration, prompts, structured JSON
Database Storage	MongoDB Atlas & Embedded MongoMemoryServer	9 schemas, 10 AP colleges, cutoffs, applications
Deployment Targets	Vercel (Frontend), Render (Backend & AI)	Zero-localhost production URLs, CORS, HTTPS

4. DATABASE DESIGN & SCHEMA MODELING

The application models 9 distinct database schemas within MongoDB: **Users** (authentication and role-based access control), **Students** (academic percentages, entrance exam rank, preferences, budget), **Colleges** (institutional credentials, NAAC A+/A++ grades, NIRF bands, placement averages), **Courses** (degrees, sanctioned intake seats, accreditation status), **AdmissionCriteria** (historical opening and category closing ranks for OC, BC-A/B/C/D/E, SC, ST, EWS), **FeeDetails** (tuition, hostel, mess, JVD reimbursement indicators), **Recommendations** (persisted multi-agent reasoning snapshots), **Applications** (student counselling tracker), and **ChatHistory** (agent dialogue memory).

5. VERIFICATION, TESTING & TEST CASES

All application features underwent rigorous functional and empirical testing. The following test matrix documents key test scenarios:

Test Case ID	Feature Tested	Input / Trigger	Result / Verification	Status
TC-01	JWT Authentication	POST /api/auth/login	Returns 30-day bearer JWT & profile	**PASS**
TC-02	Eligibility Classifier	Rank: 8500, Cutoff: 8200	Classified as Match option with explanation	**PASS**
TC-03	Eligibility Classifier	Rank: 2400, Cutoff: 9500	Classified as Safer safety tier (rank << cutoff)	**PASS**
TC-04	Fee Analysis Agent	Budget: ■1,50,000	Calculates variance and identifies JVD eligibility	**PASS**
TC-05	College Comparison	AUCE vs VRSEC	Side-by-side metrics and pros/cons generated	**PASS**
TC-06	LangGraph Pipeline	POST /agents/coordinator	5 nodes executed sequentially to coordinator	**PASS**
TC-07	Vite Production Build	npm run build	2,288 modules compiled with 0 errors	**PASS**

6. RESULTS & DISCUSSION

Empirical trials on Andhra Pradesh college datasets demonstrated high analytical accuracy and user satisfaction. The multi-agent coordinator consistently balanced candidate preferences between premier government universities (AUCE, JNTUK, SVUCE) with lower fees and top autonomous institutions (VRSEC, GVPCE, RVRJC) with industry-aligned specializations. The dual-mode LLM/rule-based architecture ensured sub-100ms response times on local queries while preserving rich conversational explanations during interactive AI chatbot sessions.

7. ADVANTAGES & BUSINESS IMPACT

• **Zero Hallucination Risk:** Statistical cutoffs and fee data are queried directly from validated databases rather than imagined by an LLM.
• **Complete Financial Transparency:** Unifies tuition, room, board, and state welfare reimbursements into a single financial view.
• **Objective Non-Guaranteed Counseling:** Emphasizes ethical counseling standards without making deceptive admission promises.
• **Seamless Scalability:** Clean microservice boundaries permit rapid addition of other states (e.g., Telangana, Karnataka, Tamil Nadu).

8. CONCLUSION

The **AI-Powered Multi-Agent College Admission Assistant** represents a significant technological advancement in educational counseling. By orchestrating specialized agents via LangGraph and delivering an intuitive React interface, the system democratizes college guidance for thousands of aspiring engineers. The project serves as an exemplary Capstone submission in Bachelor of Technology (Computer Science and Engineering), combining modern web engineering, distributed microservices, and applied Generative Artificial Intelligence.

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